Filtring in the frequency domain

Fourier serisi :

Fourier dönüşümü

$$x(t) \longleftrightarrow X(f) = \int_{-\infty}^{\infty} x(t)\, e^{-j2\pi ft}\, dt$$

DSP

$$x[n] \longleftrightarrow X[k] = \sum_{n=0}^{N-1} x[n]\, e^{-j\frac{2\pi}{N}kn}, \quad k = 0, \dots, N-1$$

yeni noktasayon

$$F[u] = \sum_{x=0}^{N-1} f[x]\, e^{-j\frac{2\pi}{M}ux}, \quad u = 0, \dots, M-1$$

$$F[u,v] = \sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)\, e^{-j\frac{2\pi}{M}ux}\, e^{-j\frac{2\pi}{N}vy} = e^{-j2\pi\left(\frac{ux}{M}+\frac{vy}{N}\right)}$$

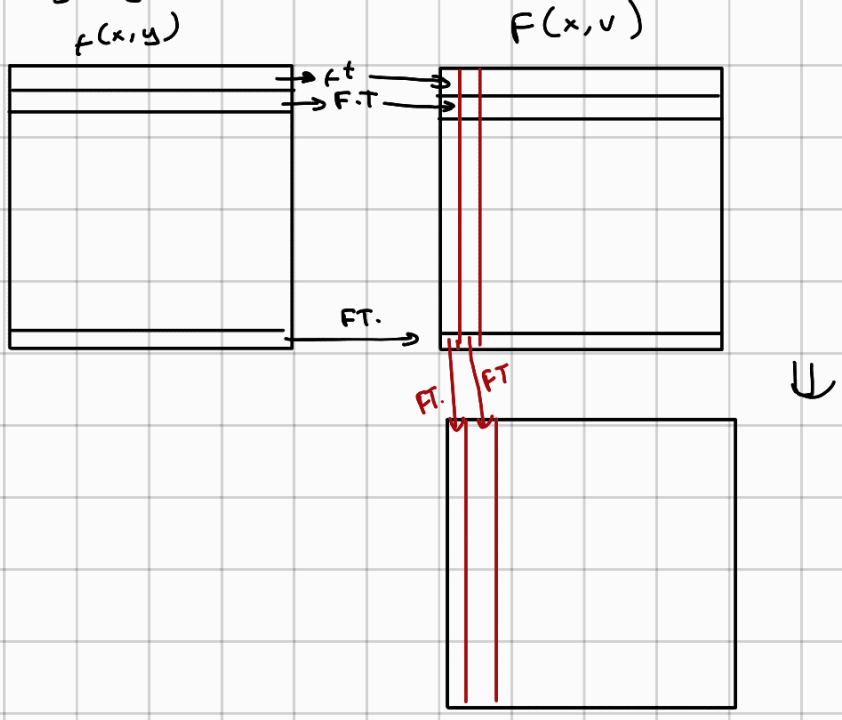
$u = 0, \dots, M-1$

$v = 0, \dots, N-1$

$$f(x,y) = \frac{1}{MN}\sum_{u=0}^{M-1}\sum_{v=0}^{N-1} F(u,v)\, e^{j\frac{2\pi}{M}ux}\, e^{j\frac{2\pi}{N}vy}$$

çok yavaş olduğu için aşağıdakini yapıyoruz

$M^2 \cdot N^2$ işlem yapılmalı



50 000 kat yazla

Bir fourier dönüşümü $N \log 2(N)$

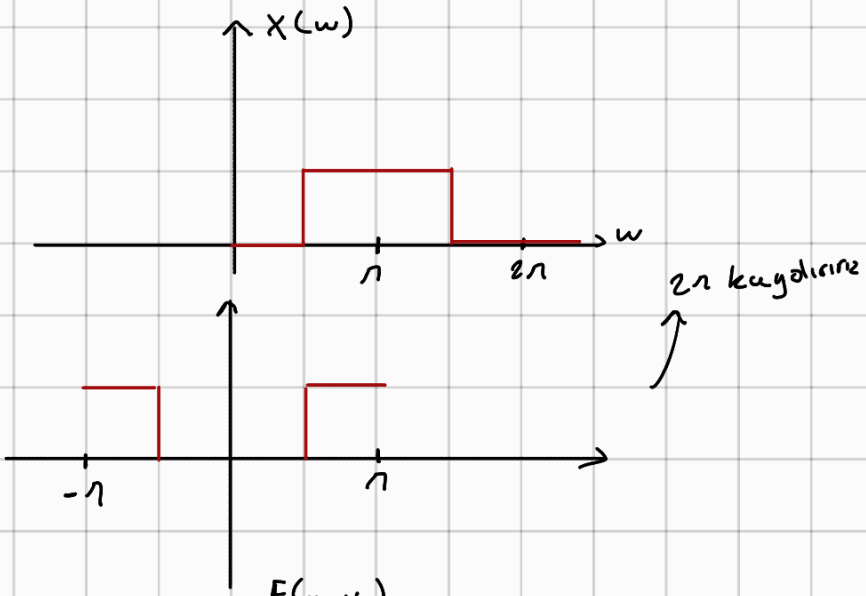
Toplam $2N^2 \log 2N$

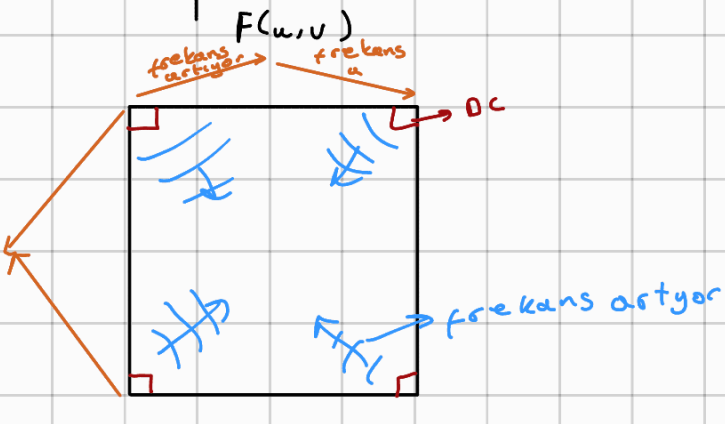
$N = 1000 \Rightarrow$ 20 milyon

$$\sum_{x=0}^{M-1}\left[\underbrace{\left[\sum_{y=0}^{N-1} f(x,y)\,e^{-j\frac{2\pi}{N}vy}\right]}_{F(x,v)} e^{-j\frac{2\pi}{M}ux}\right]$$

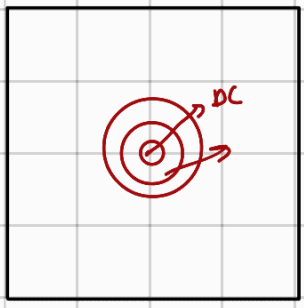
$F(u,v)$

$$f(x,y)\ \exp\left(2\pi j\left(\frac{u_0}{M}x+\frac{v_0}{N}y\right)\right) \longleftrightarrow F(u-u_0, v-v_0)$$

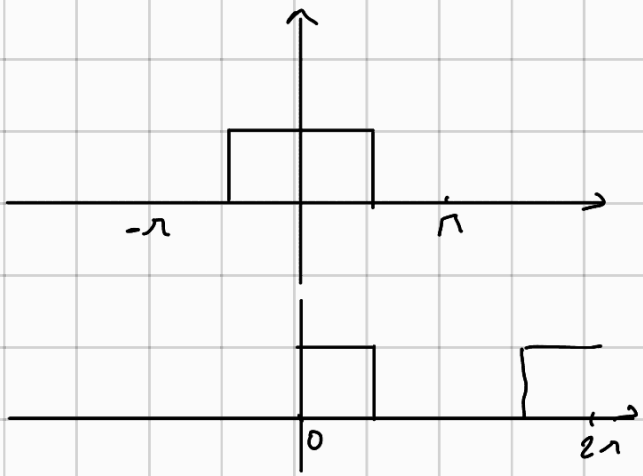




böyle yapsak



yapmak istiyoruz



Merkeze nasıl taşıyacağız?

$$f(x,y)\exp\left(2\pi j\left(\frac{u_0 x}{M}+\frac{v_0 y}{N}\right)\right) \longleftrightarrow F(u-u_0, v-v_0)$$

$u_0 = \frac{M}{2}$, $v_0 = \frac{N}{2}$

$$e^{j2\pi \frac{M}{2} x} f(x,y) \longleftrightarrow \sum_{x=0}^{M-1}\left[e^{j2\pi \frac{M}{2}\frac{x}{M}} f(x)\right] e^{-j\frac{2\pi}{M}ux} = \sum_{x=0}^{M-1} f(x)\, e^{-j\frac{2\pi}{M}\left(u-\frac{M}{2}\right)x}$$

$$\sum_{x=0}^{M-1}\left[f(x)\, e^{2\pi j \frac{u_0 x}{M}}\right] e^{-j\frac{2\pi}{M}ux} = \sum_{x=0}^{M-1} f(x)\, e^{-j\frac{2\pi}{M}(u-u_0)x} = F(u-u_0)$$

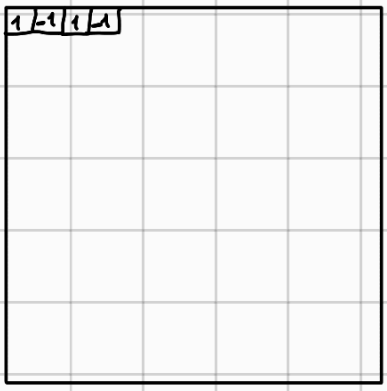
$u_0 = \frac{M}{2}$

$f(x) = e^{j2\pi \frac{M}{2} x}$

$f(x)\ (e^{j\pi})^x$

$f(x)(-1)^x \longleftrightarrow F(u-\frac{M}{2})$

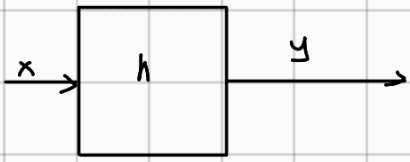
$f(x,y)(-1)^{x+y} \longleftrightarrow F\left(u-\frac{M}{2}, v-\frac{N}{2}\right)$

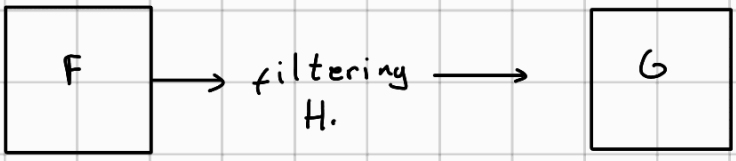


giriş reelse fourier konjüge simetriktir

if $f(x,y)$ is real , $F(u,v) = F^*(-u,-v)$

$$|F(u,v)| = |F^*(-u,-v)|$$

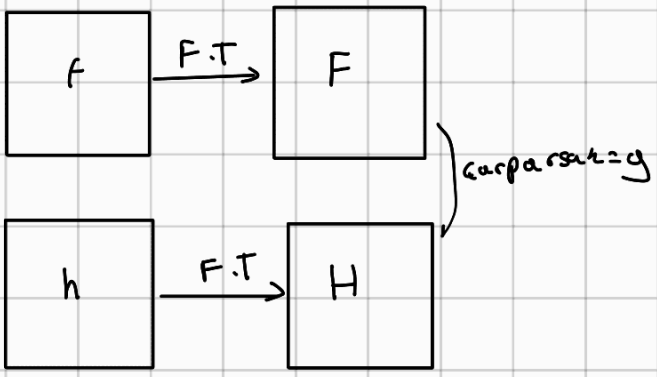




$f * h = g$ $\qquad$ $f \circ h = g$

frekans domaini

$F \cdot H = G$

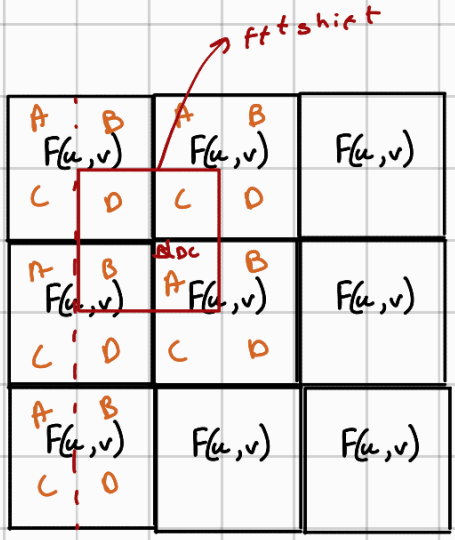


Power spectrum $\qquad P(u,v) = |F(u,v)|^2$

Average value $\qquad \bar{f}(x,y) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y) = \frac{1}{MN} F(0,0)$

\#5

Periodicity $\qquad F(u,v) = F(u + k_1 M, v + k_2 N)$



Name	Expression(s)
1) Discrete Fourier transform (DFT) of $f(x, y)$	$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M+vy/N)}$
2) Inverse discrete Fourier transform (IDFT) of $F(u, v)$	$f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi(ux/M+vy/N)}$
3) Polar representation	$F(u, v) = \lvert F(u, v)\rvert e^{j\phi(u,v)}$
4) Spectrum	$\lvert F(u, v)\rvert = \left[R^2(u, v) + I^2(u, v)\right]^{1/2}$ $R = \text{Real}(F); \quad I = \text{Imag}(F)$
5) Phase angle	$\phi(u, v) = \tan^{-1}\left[\frac{I(u, v)}{R(u, v)}\right]$
6) Power spectrum	$P(u, v) = \lvert F(u, v)\rvert^2$
7) Average value	$\bar{f}(x, y) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) = \frac{1}{MN} F(0, 0)$

TABLE 4.2
Summary of DFT definitions and corresponding expressions.

(*Continued*)

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TABLE 4.2
(*Continued*)

Name	Expression(s)
8) Periodicity (k_1 and k_2 are integers)	$F(u, v) = F(u + k_1M, v) = F(u, v + k_2N)$ $= F(u + k_1M, v + k_2N)$ $f(x, y) = f(x + k_1M, y) = f(x, y + k_2N)$ $= f(x + k_1M, y + k_2N)$
9) Convolution	$f(x, y) \star h(x, y) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m, n) h(x - m, y - n)$
10) Correlation	$f(x, y) \star h(x, y) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f^*(m, n) h(x + m, y + n)$
11) Separability	The 2-D DFT can be computed by computing 1-D DFT transforms along the rows (columns) of the image, followed by 1-D transforms along the columns (rows) of the result. See Section 4.11.1.
12) Obtaining the inverse Fourier transform using a forward transform algorithm.	$MNf^*(x, y) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F^*(u, v) e^{-j2\pi(ux/M+vy/N)}$ This equation indicates that inputting $F^*(u, v)$ into an algorithm that computes the forward transform (right side of above equation) yields $MNf^*(x, y)$. Taking the complex conjugate and dividing by MN gives the desired inverse. See Section 4.11.2.

TABLE 4.3
Summary of DFT pairs. The closed-form expressions in 12 and 13 are valid only for continuous variables. They can be used with discrete variables by sampling the closed-form, continuous expressions.

Name	DFT Pairs
1) Symmetry properties	See Table 4.1
2) Linearity	$af_1(x, y) + bf_2(x, y) \Leftrightarrow aF_1(u, v) + bF_2(u, v)$
3) Translation (general)	$f(x, y)e^{j2\pi(u_0x/M + v_0y/N)} \Leftrightarrow F(u - u_0, v - v_0)$ $f(x - x_0, y - y_0) \Leftrightarrow F(u, v)e^{-j2\pi(ux_0/M + vy_0/N)}$
4) Translation to center of the frequency rectangle, $(M/2, N/2)$	$f(x, y)(-1)^{x+y} \Leftrightarrow F(u - M/2, v - N/2)$ $f(x - M/2, y - N/2) \Leftrightarrow F(u, v)(-1)^{u+v}$
5) Rotation	$f(r, \theta + \theta_0) \Leftrightarrow F(\omega, \varphi + \theta_0)$ $x = r\cos\theta \quad y = r\sin\theta \quad u = \omega\cos\varphi \quad v = \omega\sin\varphi$
6) Convolution theorem†	$f(x, y) \star h(x, y) \Leftrightarrow F(u, v)H(u, v)$ $f(x, y)h(x, y) \Leftrightarrow F(u, v) \star H(u, v)$

(*Continued*)

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TABLE 4.3
(*Continued*)

Name	DFT Pairs
7) Correlation theorem†	$f(x, y) \star h(x, y) \Leftrightarrow F^*(u, v)H(u, v)$ $f^*(x, y)h(x, y) \Leftrightarrow F(u, v) \star H(u, v)$
8) Discrete unit impulse	$\delta(x, y) \Leftrightarrow 1$
9) Rectangle	$\text{rect}[a, b] \Leftrightarrow ab\dfrac{\sin(\pi ua)}{(\pi ua)}\dfrac{\sin(\pi vb)}{(\pi vb)}e^{-j\pi(ua+vb)}$
10) Sine	$\sin(2\pi u_0x + 2\pi v_0y) \Leftrightarrow$ $j\dfrac{1}{2}\left[\delta(u + Mu_0, v + Nv_0) - \delta(u - Mu_0, v - Nv_0)\right]$
11) Cosine	$\cos(2\pi u_0x + 2\pi v_0y) \Leftrightarrow$ $\dfrac{1}{2}\left[\delta(u + Mu_0, v + Nv_0) + \delta(u - Mu_0, v - Nv_0)\right]$
The following Fourier transform pairs are derivable only for continuous variables, denoted as before by t and z for spatial variables and by μ and ν for frequency variables. These results can be used for DFT work by sampling the continuous forms.	
12) *Differentiation* (The expressions on the right assume that $f(\pm\infty, \pm\infty) = 0$.)	$\left(\dfrac{\partial}{\partial t}\right)^m\left(\dfrac{\partial}{\partial z}\right)^n f(t, z) \Leftrightarrow (j2\pi\mu)^m(j2\pi\nu)^n F(\mu, \nu)$ $\dfrac{\partial^m f(t, z)}{\partial t^m} \Leftrightarrow (j2\pi\mu)^m F(\mu, \nu); \dfrac{\partial^n f(t, z)}{\partial z^n} \Leftrightarrow (j2\pi\nu)^n F(\mu, \nu)$
13) *Gaussian*	$A2\pi\sigma^2 e^{-2\pi^2\sigma^2(t^2+z^2)} \Leftrightarrow Ae^{-(\mu^2+\nu^2)/2\sigma^2}$ (A is a constant)

†Assumes that the functions have been extended by zero padding. Convolution and correlation are associative, commutative, and distributive.